

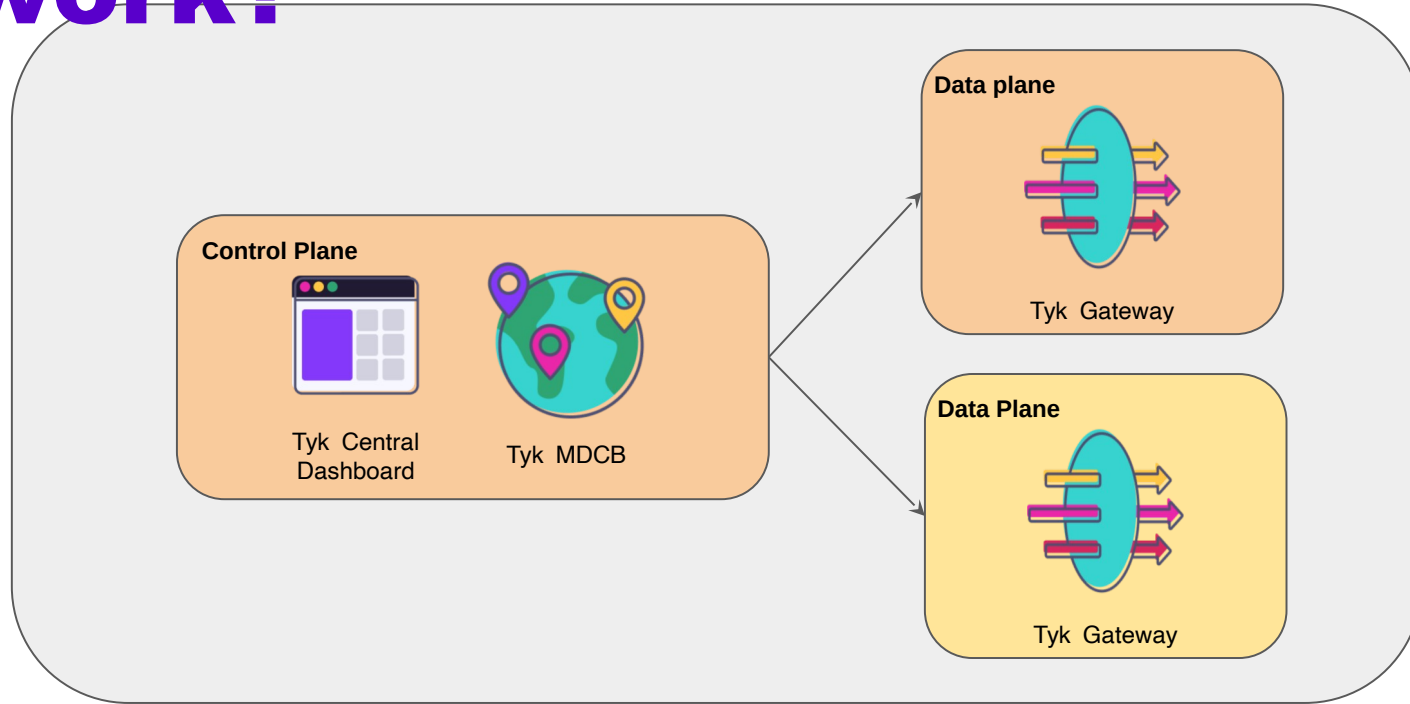
Tyk Product Components: MDCB (Multi Data Centre Bridge)

Exploring the role of each component in the Tyk Ecosystem

Tyk MDCB Overview

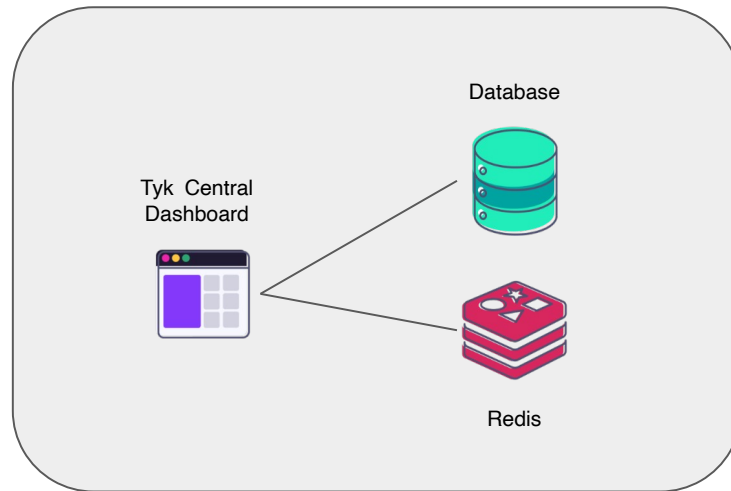
- Centralized control for distributed gateways
 - Manage geographically separated gateways from one Dashboard
- Supports multi-region and hybrid deployments
 - Ideal for global systems or compliance-restricted zones
- Worker Gateways polls MDCB periodically
 - API definitions, policies, certificates and keys are synced across regions
- Pushes analytics to Database (If MDCB is used for analytics)
 - Keeps visibility across the entire system

Tyk MDCB: How does it work?



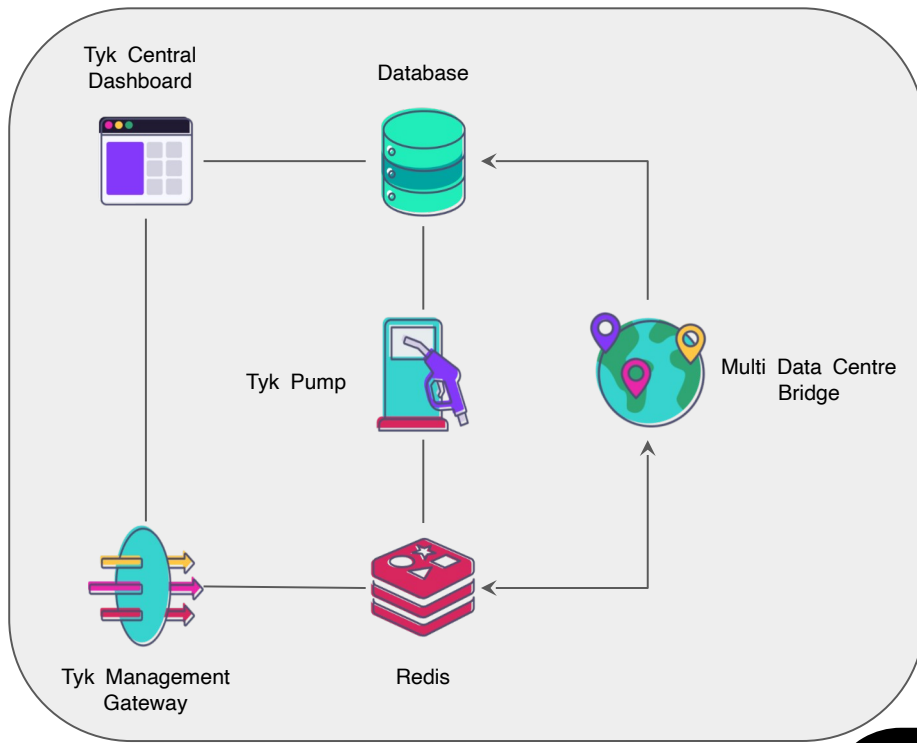
Tyk MDCB: Control Plane

- Tyk Central Dashboard
 - Control plane to manage configuration
 - Visualize analytics
- Database
 - Store analytics and configurations e.g. API definitions and policies
- Redis
 - Stores copies of all tokens across all zones and DC's
 - Used as pub/sub to emit signals on changes e.g. policy updates



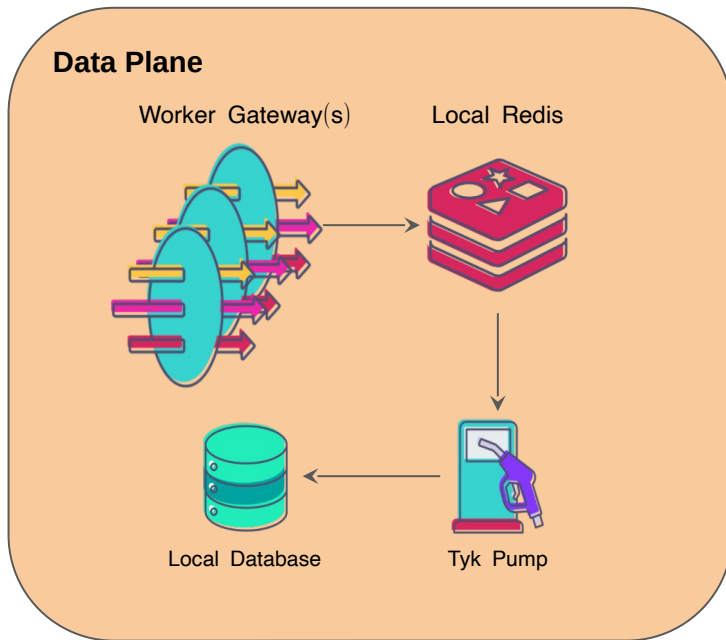
Tyk MDCB: Control Plane

- Tyk Management Gateway
 - Operates as a master gateway – does not service traffic and therefore must be isolated
 - Creates and stores API keys in Redis
- MDCB
 - Manages connections and data flow between the **Control Plane** and worker gateways in the **Data Plane**



Tyk MDCB: Data Plane

- Worker Gateway(s)
 - Handling API requests
 - Synchronizes with Control Plane
 - Grouped by Group ID
 - Operates Autonomously
- Local Redis instance
 - Worker Gateway functionality
 - Shared by all Worker Gateway(s)
- Tyk Pump
 - Localised analytics backend
 - Or Sent to MDCB as “Hybrid Pump”



Tyk MDCB: Data Plane

